

Name: Ninja Cat

Date: 12/3

Element H: Prototype Testing and Data Collection Plan

1. Durability (High Priority)

Tests: Perform static load tests to confirm the cart can support up to 500 pounds on both levels. Conduct dynamic stress tests simulating ~10,000 movement cycles to replicate five years of regular use. Use a treadmill or mechanical device to apply consistent stress.

Data Collection: Monitor deformation, wear, and failure points throughout testing and record the structural integrity after each stress testing phase.

Validation: Collaborate with materials experts to confirm that durability standards align with five-year reliability requirements.

2. Removability (High Priority)

Tests: How long does installing and removing the modification without tools take to install and remove it? Test with at least 10 users under controlled conditions to ensure consistency.

Data Collection: Record the average installation and removal time. Document user success rates (target: $\geq 80\%$ success).

Validation: Confirm practicality and ease of use through target user feedback and statistical analysis of trial outcomes.

3. Battery and Range Indicator (Medium Priority)

Tests: Conduct runtime tests with varying loads to evaluate battery performance. Monitor the accuracy of the battery indicator during operation.

Data Collection: Record runtime, charging cycles, and remaining charge after multiple trips. Evaluate indicator precision in reflecting actual battery capacity.



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Validation: Work with electrical engineers to verify the battery's energy density, longevity, and indicator accuracy.

4. Compact Design (Medium Priority)

Tests: Measure the overall dimensions of the modified cart to ensure it does not extend more than 1 foot beyond the existing frame and retains at least 90% of the original loading space. Evaluate usability during simulated loading tasks.

Data Collection: Document pre- and post-modification dimensions and space availability. Record feedback from users during simulated loading tasks.

Validation: Use focus groups to assess functionality and convenience. Confirm design compliance through dimensional analysis.

5. Push-Only Functionality (Medium Priority)

Tests: Measure the horizontal force required for forward movement using force sensors under fully loaded conditions (500 pounds). Assess user walking patterns and back posture through motion analysis during pushing tasks.

Data Collection: Record applied force, walking patterns, and user-reported strain or ease of use.

Validation: Physiotherapists review data to verify alignment with motion requirements to reduce back strain.

More:

Clamp and Attachment Mechanism

Tests: Test various clamp designs by measuring torque and reliability under stress. Position clamps close to the attachment plane to minimize torque effects.

Data Collection: Collect data on force distribution and clamp reliability over extended use.



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Validation: Confirm the mechanical reliability of the clamp design through expert analysis.

Motorization Feasibility

Tests: Evaluate trade-offs between motor weight and energy density using LiPo batteries—test force application for smooth movement under fully loaded conditions to prevent slippage.

Data Collection: Collect efficiency data from motorized prototypes.

Validation: Validate motor performance and energy density with mechanical and electrical engineers.

Expert Signature:

